

BRYCE DREYER

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EDUCATION

Oregon State University

September 2025 - Present

B.S. Electrical Engineering (projected 2028)

GPA: 3.91

Relevant Coursework

AC/DC Circuits, Circuit Design I, Digital Logic Design, Signal Processing, Vector Calculus Series

TECHNICAL SKILLS

Hardware Design / FPGA

System Verilog, Verilog, Altera Quartus, Tang Nano Toolchain

Hardware / Circuit Simulation

Mentor Graphics Modelsim, Questa, LT SPICE

High Frequency Electromagnetic Simulation

Ansys HFSS

PCB Design

Fusion PCB

Programming

Python, C++, Matlab

Computer Aided Design

Autodesk Fusion 360, Onshape

ENGINEERING EXPERIENCE

Research Assistant

October 2025 - June 2026

Collaborated with Dr. Pooya Tadayan on a novel pogo-pin design. Designed candidate pogo pins in Fusion 360, and simulated electromagnetic fields with Ansys HFSS. Created custom Python script to streamline data visualization across trials.

FIRST FRC Robotics Team Shockwave (4488)

September 2020 - June 2025

Captain and Lead Electrical of Glencoe High School's FIRST robotics team: 4488 Shockwave. Integrated electrical wiring routes and electronic placement with mechanical systems to fulfill programming needs. Worked closely with mechanical design team to create 2024 and 2025 robots.

ENGINEERING PROJECTS

CPU Branch Predictor Research Report

Summer 2026

Researched and wrote report about CPU branch prediction methodologies from basic static predictors to modern versions of the Tagged Geometric History Length Table with Statistical Corrector and Loop Predictor (TAGE-SC-L) and Convolution Neural Network (CNN).

Digital Logic Design Final Project

Spring 2026

Designed 3x3 Systolic Array matrix multiplier. Highly pipelined design provides significant clock cycle time improvement compared to serial calculation methods – approximately 8 cycles compared to 45.

Project Portfolio

<https://brycedreyer32.github.io/Website/>

Additional information about experience and projects is detailed in the above portfolio.

AWARDS AND HONORS

EECS Sophomore of the Year Nominee

Spring 2026

Eta Kappa Nu (HKN) Honors society

Spring 2026

OSU Dean's List

Winter, Spring 2026

FIRST Dean's List Semifinalist

2024